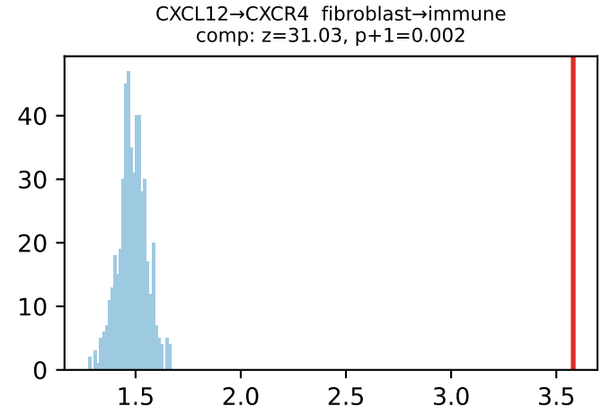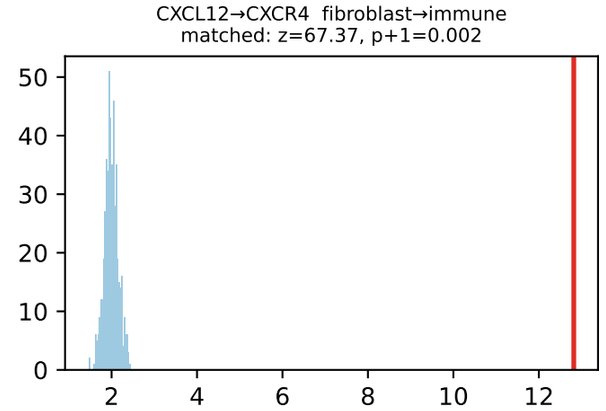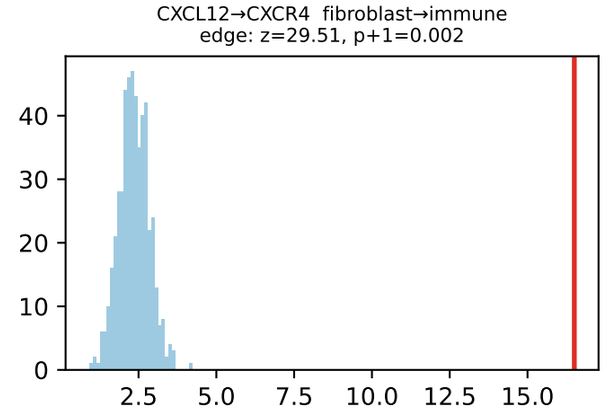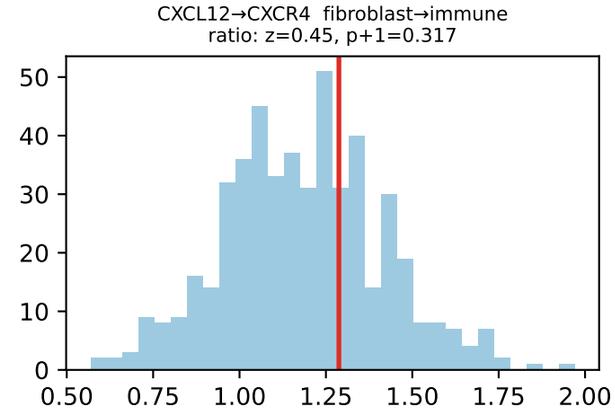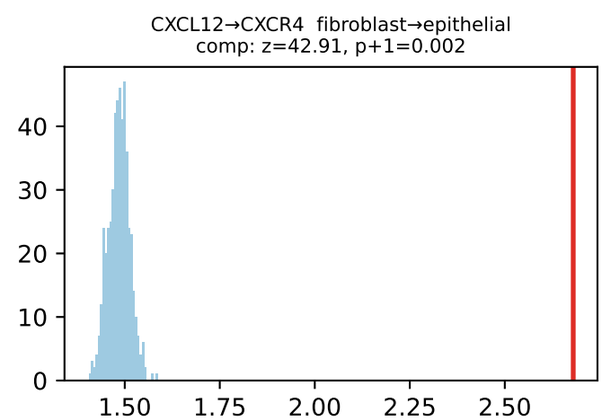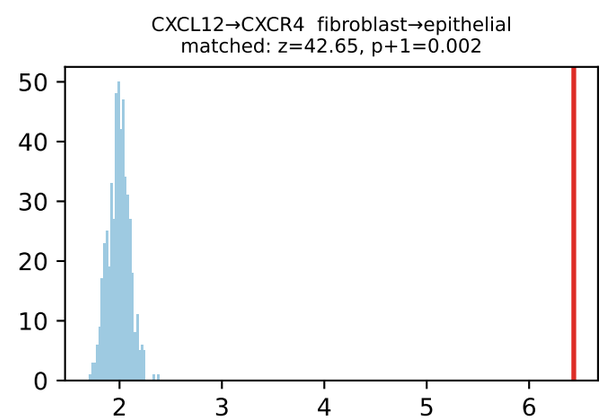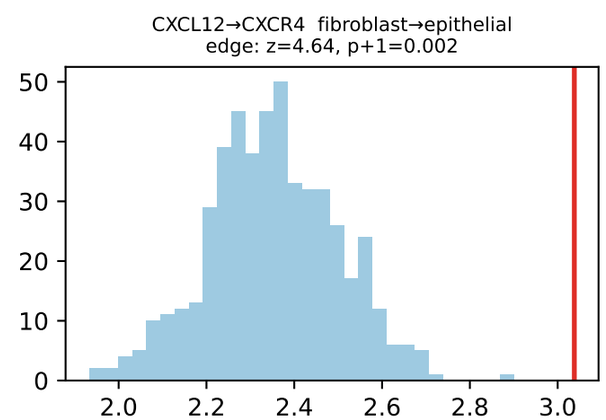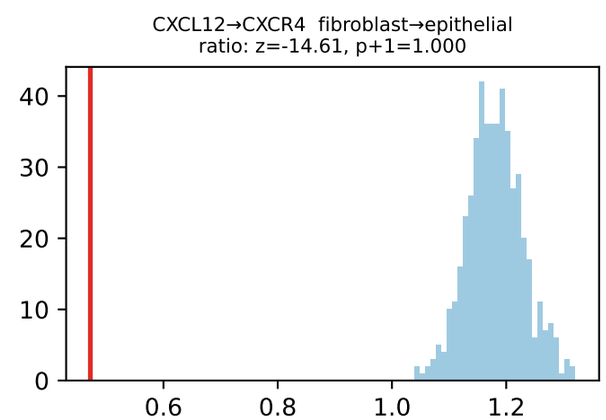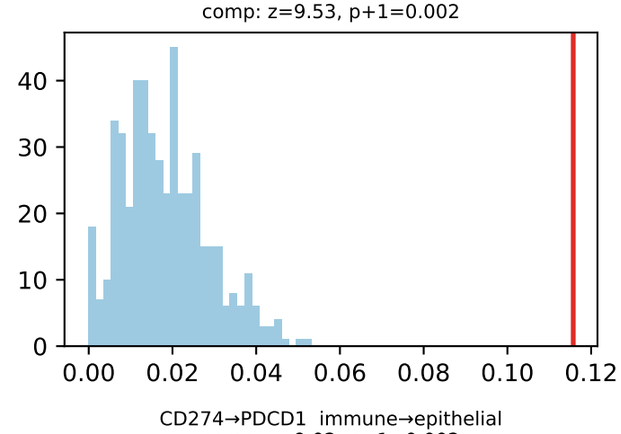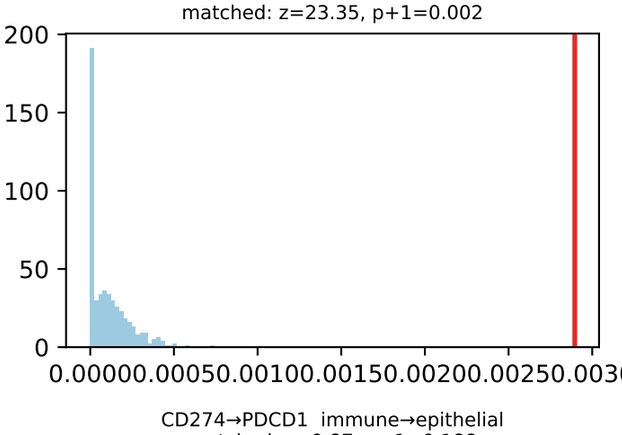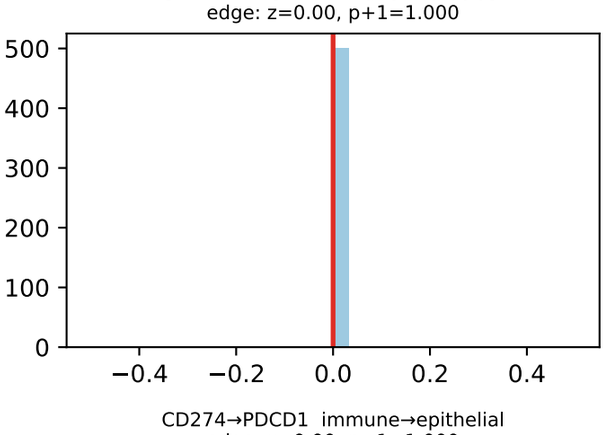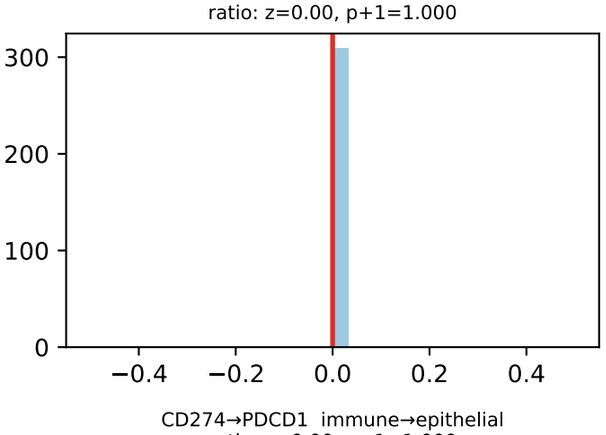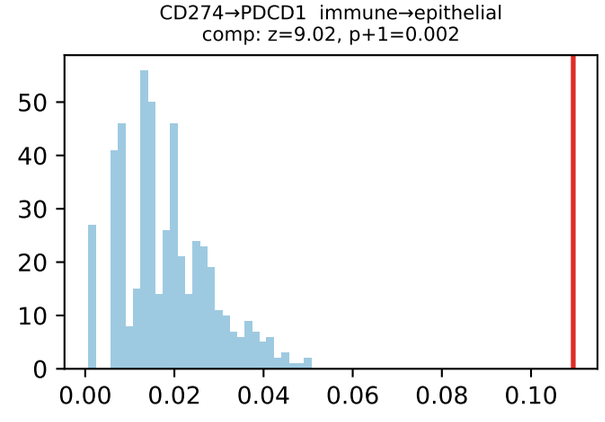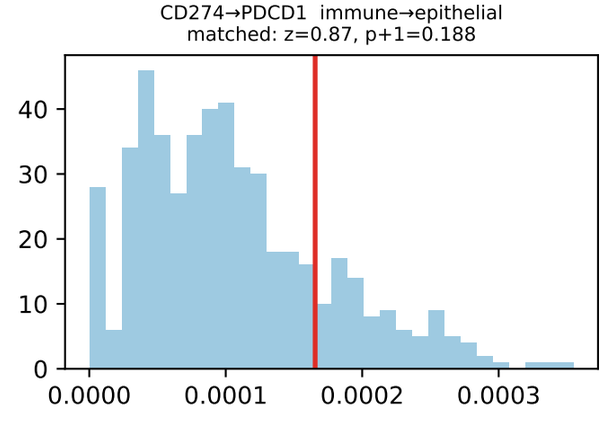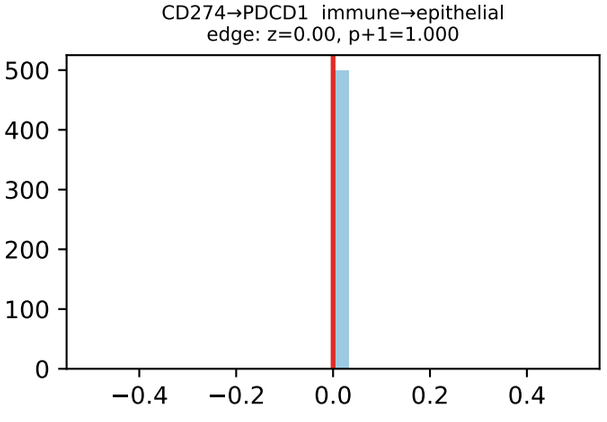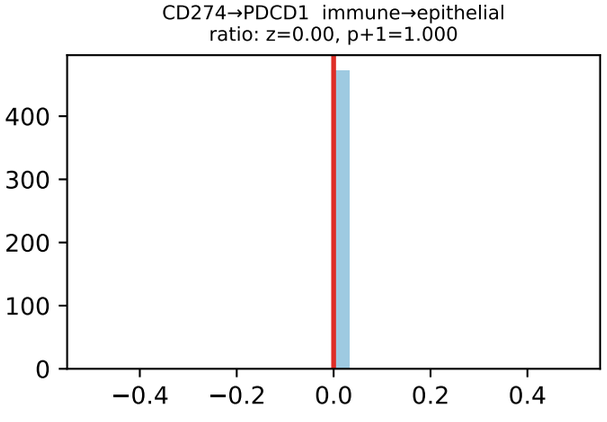Phase 1: null distributions (500 label permutations, default_rng(42)); red line = observed